MAIS PAC

IPhO 02.10: Waves, Sound and the Doppler Effect · Worked Solutions

Waves, Sound and the Doppler Effect: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

1 2 3 4 5 6 7 8 9 10

Solution 1 — A wave on a string. Return →

(a) v = √(T/μ) = √(80/(2.0×10^(-3))) = √(4.0×10^4) = 2.0×10² {m,s^(-1)}.

(b) λ = v/f = 200/120 = 1.67 {m}.

(c) v_({particle,max}) = ω A = 2π(120)(2.0×10^(-3)) = 1.5 {m,s^(-1)}, about 0.8% of the wave speed. **Check:** the two speeds are conceptually different — the crest moves at 200 {m,s^(-1)} while the string itself moves only a fraction of a metre per second, and in a sound wave the ratio is even more extreme.

Solution 2 — Harmonics of a string. Return →

(a) f_1 = v/(2L) gives v = 2Lf_1 = 2(0.65)(196) = 255 {m,s^(-1)}.

(b) f_n = nf_1, so 392 {Hz}, 588 {Hz} and 784 {Hz} — an octave, then a fifth above the octave, then two octaves.

(c) Pressing at the midpoint halves the vibrating length to 0.325 {m}, so the fundamental becomes v/(2×0.325) = 392 {Hz}, exactly twice the open-string fundamental: an octave higher. **Check:** the harmonic series of the shortened string is 392, 784,, which overlaps the second harmonic of the full string — the reason a guitar's harmonic at the 12th fret sounds the same as the stopped note.

Solution 3 — Sound intensity and decibels. Return →

(a) I = (P)/(4π r²) = (2.0)/(4π(3.0)²) = 1.77×10^(-2) {W,m^(-2)}.

(b) L = 10log _{10}(1.77×10^(-2))/(10^(-12)) = 10log _{10}(1.77×10^(10)) = 102 {dB}.

(c) A drop of 20 {dB} is a factor of 10^(-2) in intensity, and since I∝1/r² it means r has grown by a factor of 10: r = 30 {m}, where I = 1.77×10^(-4) {W,m^(-2)}. **Check:** 10log _{10}(1.77×10^6)≈62 {dB}, which is 102-40, consistent with the inverse square law over a factor of 10 in distance (two factors of 2 in amplitude, four in intensity).

Solution 4 — Beats. Return →

(a) f_({beat}) = |444-440| = 4.0 {Hz}.

(b) The listener hears a 442 {Hz} tone (the mean) whose loudness rises and falls four times a second. Adding a third fork at 442 {Hz} gives beats at 2.0 {Hz} against each of the others, a more complex and slower pulsation rather than a single clean beat.

(c) The tuner sounds the reference and the string together and slackens or tightens the string until the beats slow down and vanish, which happens when the frequencies coincide exactly. Beats turn a frequency difference into a **time** interval, which the ear judges far more accurately than pitch. **Check:** the beat frequency is the difference, not the sum: two waves of 440 and 444 {Hz} reinforce twice per cycle of the difference, giving four maxima per second.

Solution 5 — A pipe closed at one end. Return →

(a) A closed pipe resonates at odd multiples of the fundamental. Here 960/320 = 3, so the two resonances are the first and third harmonics of a pipe whose fundamental is 320 {Hz} — indeed 3×320 = 960.

(b) L = (v)/(4f_1) = (340)/(4(320)) = 0.266 {m}.

(c) The series is f_n = (2n-1)f_1 = 320, 960, 1600, 2240 {Hz}. There is no resonance at 640 {Hz} because that would require a wavelength of 340/640 = 0.53 {m}, and L = λ/2 puts an antinode at the closed end — impossible, since the air cannot move there. **Check:** 1600 = 5f_1, and 5×0.266 = 1.33 {m} = 5λ/4 with λ = 340/1600 = 0.2125 {m}, as required.

Solution 6 — The Doppler effect for an ambulance. Return →

(a) f_({app}) = f(v)/(v-v_s) = 700×(340)/(315) = 756 {Hz}.

(b) f_({rec}) = 700×(340)/(365) = 652 {Hz}.

(c) The drop is 104 {Hz}, about 14% of 700 {Hz}. The change is abrupt because the sign of v_s in the formula flips as the ambulance passes: the frequency is not a continuous function of position at the moment of closest approach, although the **received** signal is continuous — the sound emitted at the moment of passing arrives with an intermediate frequency. **Check:** 756/652 = 1.16, and the two shifts are roughly ± v_s/v = ± 7%, consistent with the small-v approximation.

Solution 7 — Moving source and moving observer. Return →

(a) Both effects increase the frequency:

$$f_{obs} = f,(v+v_o)/(v-v_s) = 500\times(340+20)/(340-30) = 500\times(360)/(310) = 581\ \text{Hz}. \quad (1)$$

(b) With a stationary source and an observer at 50 {m,s^(-1)}:

$$f = 500\times(340+50)/(340) = 574\ \text{Hz}. \quad (2)$$

(c) The answers differ by 7 {Hz} because sound travels through air, which singles out a rest frame: a moving source changes the crest spacing in the air, while a moving observer only changes how often the crests are met. Light has no such frame, and the two cases give identical shifts, which is the starting point of special relativity. **Check:** at speeds much smaller than v both formulas give f(1 + (v_s+v_o)/v) = 500(1+50/340) = 574 {Hz}, so the difference is a second-order effect.

Solution 8 — Interference from two speakers. Return →

(a) λ = v/f = 340/200 = 1.7 {m}.

(b) Directly in front of one speaker, the distances are 4.0 {m} and √(4.0²+3.0²) = 5.0 {m}, so Δ = 1.0 {m}, which is 0.59λ — neither a maximum nor a minimum, but close to the minimum at λ/2 = 0.85 {m}.

(c) Walking far to either side, the path difference tends to the separation 3.0 {m} = 1.76λ. Maxima occur where Δ = nλ, i.e. Δ = 0, 1.7, 3.4 {m}; since Δ runs from -1.76λ to +1.76λ, the values n = -1, 0, 1 are crossed, giving three maxima. **Check:** midway between the speakers Δ = 0 always, so the centre line is a maximum whatever the frequency.

Solution 9 — Standing waves and resonance in a tube. Return →

(a) Open at both ends: f_n = nv/(2L) = n(340)/(1.0) = 340n {Hz}, so 340, 680 and 1020 {Hz}.

(b) Closed at one end: f_n = (2n-1)v/(4L) = (2n-1)(170) {Hz}, so 170, 510 and 850 {Hz}.

(c) The fundamental of the closed tube is v/(4L), exactly half that of the open tube, v/(2L): closing one end allows a quarter wavelength instead of a half, so the same length holds a wave of twice the wavelength and half the frequency. **Check:** the closed tube's second resonance, 510 {Hz}, is three times its fundamental, and the open tube's second, 680 {Hz}, is twice — the even harmonic is missing exactly because of the node at the closed end.

Solution 10 — A wave on a string with a boundary. Return →

(a) The boundary is a change of impedance, and part of the pulse is transmitted into the heavier string and part is reflected. Since the heavier string is "harder to move", the boundary behaves partly like a fixed end: the reflected pulse is **inverted**.

(b) With the lighter string second, the boundary behaves partly like a free end and the reflected pulse is **upright**. Mathematically the reflected amplitude is proportional to Z_1 - Z_2 with Z = √(Tμ), so the sign flips when the order of the strings is swapped.

(c) A fixed end is the limit of an infinitely heavy second string: total reflection with inversion. A free end is the limit of a vanishingly light one: total reflection without inversion. Real boundaries fall in between, transmitting a fraction of the energy. **Check:** in both limits the transmitted amplitude goes to zero, but the phase of the reflection is opposite — and that phase difference is exactly what decides whether a standing wave has a node or an antinode at the boundary.